Final Course Project Guidelines

Comprehensive Data Analysis & Business Reporting

Final Report Deadline: 11/05/2026 — **Total Marks:** 100

Overview

This semester project requires you to apply the statistical concepts learned in class to a real-world dataset. The project is divided into three main tasks: **Task 1 (Data Collection & Descriptive Statistics)**, **Task 2 (Hypothesis Testing)**, and **Task 3 (Correlation, Regression, & Time Series)**.

Below are the detailed instructions for executing each phase. Ultimately, you will combine the results of all three tasks into a single, professional final report.

Task 1: Data Collection, Cleaning, & Descriptive Statistics

1. Data Collection & Description

- **Select a Dataset:** Find a real-world dataset relevant to a business, economic, or social problem (e.g., from Kaggle, World Bank, Yahoo Finance, or primary survey data).
- **Dataset Requirements:** Your dataset must have a minimum of 100 observations (rows) and at least 5 variables (columns), including a mix of *quantitative* (continuous/discrete) and *categorical* variables.
- **Data Dictionary:** Provide a brief description of what each variable represents and its unit of measurement.

2. Data Cleaning

- Identify and handle missing values (e.g., imputation or removal).
- Identify any severe outliers and briefly explain how you handled them.

3. Descriptive Statistics & Visualizations

- **Central Tendency & Dispersion:** Calculate the Mean, Median, Mode, Standard Deviation, Variance, and Range for your key quantitative variables.
- **Frequencies:** Calculate frequency distributions and percentages for your categorical variables.
- **Visualizations:** Create appropriate charts to visualize your data distribution. You must include at least:
  - One **Histogram** to show the distribution of a continuous variable.
  - One **Box Plot** to show the spread and outliers.
  - One **Bar Chart** or **Pie Chart** for categorical data.
- **Interpretation:** Write a brief summary explaining the "story" your descriptive statistics and graphs are telling.

Task 2: Hypothesis Testing

1. Formulating Business Questions

- Based on your dataset, formulate at least **two specific business questions** that can be answered using inferential statistics.
- Clearly state the **Null Hypothesis** (H_0) and the **Alternative Hypothesis** (H_1) for each question.

2. Selecting and Running the Tests

- Choose the appropriate statistical test for your hypotheses. You must run at least two different types of tests from the following list:
  - One-Sample or Two-Sample T-Test (Independent or Paired).
  - Analysis of Variance (ANOVA) for comparing three or more groups.
  - Chi-Square Test of Independence for categorical variables.
- Conduct the tests using your chosen software (Excel, SPSS, R, Python) using a significance level of 5% ($\alpha = 0.05$).

3. Reporting and Interpretation

- **Report the Results:** State the Test Statistic (e.g., t, F, χ^2), the Degrees of Freedom (df), and the P-value.
- **Statistical Conclusion:** State clearly whether you Reject or Fail to Reject the Null Hypothesis.
- **Business Interpretation:** What does this statistical conclusion mean in practical, real-world business terms?

Task 3: Correlation, Regression, & Time Series Analysis

1. Correlation Analysis

- Generate a correlation matrix for your quantitative variables.
- Identify the strength and direction of the relationship between your key variables.
- **Interpretation:** Discuss which variables are strongly positively or negatively correlated. Are there any surprising relationships?

2. Linear Regression Analysis

- Select one **Dependent Variable** (Y) and one or more **Independent Variables** (X) based on your hypothesis.
- Run a Simple or Multiple Linear Regression model.
- **Reporting Requirements:**
  - Present the **Regression Equation** ($\hat{y} = b_0 + b_1 x$).
  - Interpret the **Coefficient of Determination** (R^2): How much variance in Y is explained by X?

- Interpret the **Slope Coefficient(s)**: What is the impact of a one-unit change in X on Y?
- Discuss the **Significance (P-value)**: Is the relationship statistically significant at the 5% level ($\alpha = 0.05$)?

3. Time Series Analysis (Conditional)

- *If your dataset includes a time variable (e.g., dates, years, months):*
  - Create a **Time Series Plot** to visualize the trend of your main variable over time.
  - Comment on any visible trends, seasonality, or cyclical patterns.
- *If your dataset is Cross-Sectional (no time variable):*
  - Create a detailed **Scatter Plot** showing the relationship between your Independent and Dependent variables with a trendline added.

The Final Report Structure

You must now combine the work from Task 1, Task 2, and Task 3 into a single, professional document.

1. **Title Page:** Project Title, Group Members, Course Name.
2. **Executive Summary:** A 1-page summary of the entire project (Problem, Method, Key Findings).
3. **Introduction:** Background of the study, research questions, and data source description.
4. **Descriptive Analysis:** Summary statistics and initial visualizations (from Task 1).
5. **Hypothesis Testing:** Hypotheses, test results, and findings (from Task 2).
6. **Predictive & Trend Analysis:** Correlation, Regression, and Time Series results (from Task 3).
7. **Conclusion & Recommendations:** What do these results imply for business or decision-making? What should be done based on your findings?
8. **Appendix:** Snapshots of raw software output (optional).

Submission Requirements

Please submit the following items in a single ZIP folder or as separate attachments on **Google Classroom (GCR)**.

Important: Only **one group member** may upload the report on GCR.

1. **The Final Report:** MS Word (.docx) or PDF. Times New Roman, 12pt, 1.5 spacing. Ensure all tables and graphs are properly labeled (e.g., *Figure 1: Sales vs. Advertising*).
2. **The Dataset:** The clean Excel (.xlsx) or CSV file used for the analysis.
3. **The Analysis File:** The output file showing your work (Excel with Toolpak output sheets, SPSS .spv file, or Python/R scripts).

Grading Criteria

- **Data Prep & Descriptive Stats (20%):** Quality of data cleaning and appropriate use of summary statistics and initial charts.
- **Hypothesis Testing (20%):** Correct selection, execution, and reporting of statistical tests.
- **Regression & Modeling (20%):** Correct calculation and interpretation of Correlation, Regression coefficients, and R^2.
- **Business Interpretation (30%):** Ability to explain what the statistical numbers mean in plain English and provide actionable business recommendations.
- **Formatting & Professionalism (10%):** Clarity of visualizations and overall organization of the final document.